Rev A | 2026-10-24 | UHD-BOTTLE-AG

ASSEMBLY GUIDE

Smart Water Bottle Cap

Sensor cap for a Nalgene 32 oz Wide Mouth bottle: ESP32-C3 (Wi-Fi + BLE), VL53L4CD time-of-flight water level, Adafruit 381 DS18B20 temperature probe, LP5024 with 8 RGB LEDs, TPS63020 3V3, MCP73831 charger and MAX17048 fuel gauge on a 1000 mAh LP503450 cell, IP67 USB-C, in a printed PETG cap sealed with AS568 O-rings.



Exploded along each joint's axis in fastener-stack order (2 mates, 4 hardware parts).

STEPS	MATES	HARDWARE	PARTS
6	2	4	79
mechanical and wiring	from assemble()	screws, spacers, standoffs, nuts	40 BOM lines

HOW THIS GUIDE IS MADE

Steps, parts and hardware come from the UHD model: mates and fastener stacks from `assemble()`, wiring from `wiringChecklist`. The coverage table at the end lists where every mate, fastener and wire appears.

[A] Assumption: the model marks this value as an assumption (assumption trait)

[D] Derived: computed from model values; formula in the provenance appendix

Unmarked values are cited from the part's sources (sources.json) or are design values of the model.

01 Before you begin

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TOOLS

- Soldering iron, solder, heat-shrink

Derived from the fasteners in the model: hex keys from each screw's stated socket or the ISO 4762 socket size for its thread, drivers and spanners from each nut's and standoff's across-flats.

READING THE FIGURES

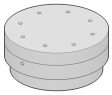
Already assembled
Fitted in this step
Hardware for this step
------ Assembly axis

Parts are drawn exploded along each joint's axis in stacking order, from the model's fastener stacks and mate gaps.

NOTE Each step's torque line gives the tightening torque of the part turned, from the harness's fastener stack. 1 of them are **assumptions** (marked A): no frame, motor or DJI source states a torque, so they are derated from supplier tables for the screw grade into steel. Threadlocker is only advised where a harness says so (steel into aluminium).

02 Kit contents

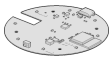
DOCS/SMART-BOTTLE-BOM.JSON



×1

Bottle cap housing (with its parts fitted)

bottle-cap-enclosure



×1

Bottle cap board (assembled)

bottle-cap-board



×1

Adafruit 381 waterproof DS18B20 temperature probe

adafruit-381-ds18b20-probe



×1

AS568-010 O-ring, silicone 70A, FDA grade

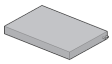
as568-010-silicone



×1

AS568-138 O-ring, silicone 70A, FDA grade

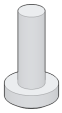
as568-138-silicone



×1

LP503450 1000 mAh 3.7 V LiPo with PCM

lipol-lp503450



×4

M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless

iso-7045-m2x6-a2



×1

Nalgene 32oz Wide Mouth Bottle, Tritan Renew

nalgene-32oz-wide-mouth



×1

O-ring AS568-040, silicone 70A (72.75 × 1.78 mm)

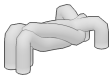
as568-040-silicone



×1

Cell leads

bottle-battery-leads



×1

Probe cable

bottle-probe-cable

Quantities from the generated BOM; a circuit board is one item, assembled by the fab (its parts are in the BOM, not at the bench); a made part with parts of its own (a housing with its inserts set) is one item; drawings are the parts' CAD artifacts, not to scale.

03 Hardware

FASTENER STACKS

Every screw, spacer, standoff and nut is placed by a harness's `fastenerStack`. The sections below are drawn from those positions and the mate gaps: one hole of each joint, to one scale, millimetres from the structure face.

HARNESS	PARTS PLACED	TORQUE	USED IN
Board hardware (M2) joint board_on_bosses	4 × M2 x 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless	0.15 N·m[A] M2 x 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless	Step 1

Board hardware (M2) bottle-board-hardware



1 Fit the board on the cap's bosses

mates: board_on_bosses



(A) ×1
Bottle cap board



(B) ×4
M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless

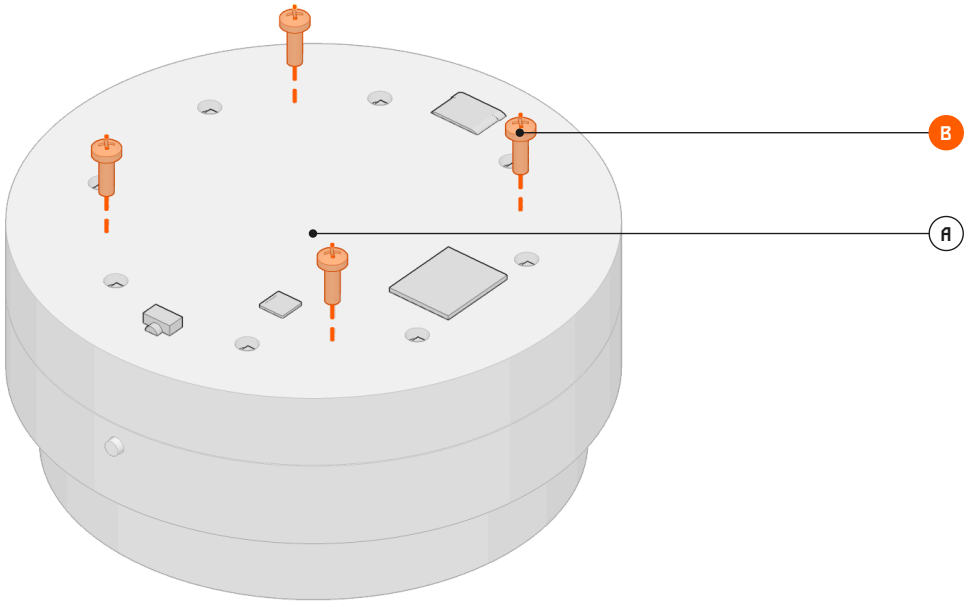


Figure 1 Fit the board on the cap's bosses: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Push the button plunger (printed with the cap) into its bore in the side wall from inside, flange inward; the board's switch holds it once the board is in.
2. Seat the Bottle cap board onto the board bosses (4 × M2 inserts).
3. Drive 4 × M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless from above, head down.

Torque 0.15 N·m[A] M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless **Step** 1 of 6

TIP The cap comes from its own unit with the four heat-set inserts set and the light pipes filled; nothing is fitted in its grooves yet.

CAUTION The receptacle's tongue faces the USB-C port: check its gasket sits square to the wall before tightening.

NOTE The inserts come set in the cap's bosses (cap unit). Tighten the screws snug, crosswise. — bottle-board-hardware

2 Fit the temperature probe and solder its cable

links: probe_wires, probe_pads



(A) ×1
Adafruit 381 waterproof DS18B20 temperature probe



(B) ×1
AS568-010 O-ring, silicone 70A, FDA grade



(C) ×1
Probe cable

Figure 2 Pad-to-pad connections from the stored links (wiringChecklist); crossovers in accent.

1. Connect Probe cable leaves the probe, Probe cable on the probe pads as in the table (6 connections).

LINK	FROM	TO	TERMINATION	DRC
Probe cable leaves the probe	probe **VDD · DQ · GND**	Probe cable **P1 · P2 · P3**	integral	CONFIGURED
Probe cable on the probe pads	Probe cable **P1 · P2 · P3**	board **1 · 2 · 3**	solder_leads	CONFIGURED

Step 2 of 6

TIP Grease the AS568-010 O-ring lightly with silicone grease and push the probe up through the floor seal until the heat shrink meets the boss.

CAUTION Only the stainless tube enters the bottle; keep the PVC cable inside the cap (OQ-B5).

NOTE Cut the probe's 91 cm cable to length, strip the jacket back 10 mm and the wires 2 mm, and solder VDD (red), DQ (yellow/white) and GND (black/blue) to the pads marked on the board. Only the stainless tube goes into the bottle; the PVC jacket stays in the dry compartment (OQ-B5). — bottle-probe-cable

3 Fit the cell and solder its leads

links: cell_leads, bat_pads



(A) ×1
LP503450 1000 mAh 3.7 V LiPo with PCM



(B) ×1
Cell leads

Figure 3 Pad-to-pad connections from the stored links (wiringChecklist); crossovers in accent.

1. Connect Cell leads leave the cell, Cell leads on the BAT pads as in the table (4 connections).

LINK	FROM	TO	TERMINATION	DRC
Cell leads leave the cell	cell **+ · -**	Cell leads **P1 · P2**	integral	CONFIGURED
Cell leads on the BAT pads	Cell leads **P1 · P2**	board **1 · 2**	solder_leads	CONFIGURED

Step 3 of 6

WARNING Red to BAT+, black to BAT-. Never short the leads: the cell's protection board is the only protection. Tape the cell to the module shield with 0.5 mm foam tape.

NOTE The cell's own leads: trim to length, tin, and solder red to BAT+ and black to BAT-. Never short them: the PCM is the only protection. — bottle-battery-leads

4 Fit the lid and lip O-rings



Ⓐ ×1

O-ring AS568-040, silicone 70A (72.75 × 1.78 mm)



Ⓑ ×1

AS568-138 O-ring, silicone 70A, FDA grade

1. Fit the O-ring AS568-040, silicone 70A (72.75 × 1.78 mm).
2. Fit the AS568-138 O-ring, silicone 70A, FDA grade.

Step 4 of 6

TIP Seat the AS568-040 in the collar's groove above the lugs and the AS568-138 in the floor's groove without twisting them; grease the AS568-040 lightly with silicone grease.

5 Close the lid

1. Push the lid down over the collar with each of its three slots' open ends on a lug (the wide slot on the wide lug, at the USB-C port's side: the lid goes on only that way), until it seats on the rim, then turn it 30° clockwise (seen from above) until the lugs stop at the slots' ends: the light pipes then stand over the LEDs.

Step 5 of 6

TIP The skirt's lead-in rides over the greased AS568-040; no tool and no screws. To open, turn back 30° and lift.

6 Screw the cap onto the bottle

mates: cap_on_bottle



Ⓐ ×1

Nalgene 32oz Wide Mouth Bottle, Tritan Renew



Figure 4 Screw the cap onto the bottle: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Seat the Nalgene 32oz Wide Mouth Bottle, Tritan Renew onto the bottle lip seal (AS568-138).

Step 6 of 6

⚠ **CAUTION** Bring-up and the bench test run with the cap off the bottle. Close to Nalgene's 4.5-6.3 N·m (assumption from the lab line) and no tighter.

10 Coverage

ASSEMBLE() · WIRINGCHECKLIST

Every rigid mate, every hardware placement and every electrical link in the model, and the step that covers it. The verifier fails the guide if any row is not covered.

MATES

MATE	INTERFACES	HARNESS	STEP
board_on_bosses	cap.board_mount ↔ board.mount	board_hardware	1 · Fit the board on the cap's bosses
cap_on_bottle	cap.lip_seal ↔ bottle.lip_seal	—	6 · Screw the cap onto the bottle

HARDWARE PLACEMENTS

HARNESS / CHILD	PARTS	JOINT	STEP
board_hardware/screws	4 × M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless	board_on_bosses	1 · Fit the board on the cap's bosses

WIRING LINKS

LINK	NAME	HARNESS	STEP
cell_leads	Cell leads leave the cell	—	3 · Fit the cell and solder its leads
bat_pads	Cell leads on the BAT pads	—	3 · Fit the cell and solder its leads
probe_wires	Probe cable leaves the probe	—	2 · Fit the temperature probe and solder its cable
probe_pads	Probe cable on the probe pads	—	2 · Fit the temperature probe and solder its cable